



ANSI/IES **LS-2-20(R2023)**

LIGHTING SCIENCE: CONCEPTS AND LANGUAGE OF LIGHTING

AN AMERICAN NATIONAL STANDARD



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has been approved by the IES.
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should be directed to the IES.

Prepared by
The IES Nomenclature Committee



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Approved by the IES Standards Committee October 11, 2023 as a Transaction of the Illuminating Engineering Society.

Approved January 19, 2024 as an American National Standard.

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Published by the Illuminating Engineering Society, 120 Wall Street, New York, New York 10005.

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Printed in the United States of America.

ISBN 978-0-87995-472-7

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1.0 Introduction and Scope

1.1 Introduction

Lighting's language fulfills the need to describe, specify, and evaluate luminous environments. Like any language, it is based on concepts and vocabulary. The concepts result from a consideration of the nature of light, vision, and architecture. The vocabulary results from the need for clarity, specificity, and precision. The structure of lighting's concepts is an inverted pyramid: a very few fundamental ideas are identified and described, and from these more-complex concepts are constructed. Simpler concepts form the constituents of the more complex ones required to unambiguously specify luminous quantities or the photometric behavior of materials. In this Lighting Science document, the fundamental or most basic concepts are described first, many of which have their roots in the work of Johann Lambert and André Blondel.¹ These are followed by more-complex or derived concepts.

1.2 Scope

Only the most important quantities and units used in lighting design and illuminating engineering that relate directly to optical radiation, light, and vision are described and defined in this document. The technical words associated with lighting equipment, photometry, lighting calculations, color, and daylighting are defined in other documents in the IES Library, and they rely on an understanding of the material presented in this document. (Refer to *ANSI/IES LS-1-22, Lighting Science: Nomenclature*

and *Definitions for Illuminating Engineering*² for the locations of the definition of specific words.)

A full nomenclature and many more derived and specialized quantities are also described in the International Lighting Vocabulary, established by the International Commission on Illumination (CIE) and published jointly with the International Electrotechnical Commission.³ More than 900 technical definitions of concepts and quantities are given in English, French, German, and Russian.

2.0 Language and Concepts

2.1 General Words

Lighting's conceptual vocabulary adopts words found in common usage and gives them a special, technical meaning. Precision in describing concepts makes this necessary.

radiant energy: This is the general term for energy propagated by radiation through a vacuum or a material, in distinction to energy transported by conduction or convection. The term is used when no particular model of energy transport is implied or when any wavelength or frequency can be involved. Three specific types of radiant energy include:

- **electromagnetic radiation:** In some cases, it is necessary or convenient to imply one of the two physical models of radiative energy transport: electromagnetic waves or photons. (Refer to *ANSI/*

SIDEBAR

Descriptive words are an important part of Lighting's vocabulary. In English, lighting concept names often derive from a stem word, usually a verb, to which suffixes are added, abiding by the following general customs of usage:

-ance added to the verb creates a noun related to an action.
This is usually the noun of quantity.

-ive or **-ing** added to the verb creates an adjective of nature that describes having the character of an action.

-ivity added to the verb, or **-ity** added to an adjective, creates a noun of abstraction, giving a name to the active property.

-tion added to the verb creates a noun of state or condition.

An example of this vocabulary construction using the word "reflect" is:

reflect. Verb; to bounce off.

reflectance. Noun of quantity; the amount of reflecting.

reflective. Adjective of nature; able to reflect.

reflectivity. Noun of abstraction; the property of being reflected.

reflection. Noun of state or condition; being reflected.

IES LS-3-20(R2023), *Lighting Science: Physics and Optics of Radiant Power* for additional information.⁴⁾ Electromagnetic radiation is radiant energy propagated in a way consistent with the model of electric and magnetic waves. For example, radiant energy moving through glass or plastic optical components is conveniently described using the electromagnetic wave model.

- **photon radiation:** This is radiant energy propagated in a way consistent with the quantum model. The energy transport within a light emitting diode is best described with the photon model.
- **optical radiation:** This is energy propagated by radiation when its wavelengths are between 100 nm and 10,000 nm, that is, radiant energy with wavelengths limited to the ultraviolet, visible, and infrared. No particular model of energy transport is implied with this term.

radiant power: In electrical engineering, the distinction between energy and power is essential, and is clear from the different uses and meanings of kilowatt (power) and kilowatt-hour (energy). This distinction between energy and power is also made when dealing with radiant quantities: radiant power is the time-rate of flow of radiant energy. It is customary to refer to radiant power as *radiant flux*; “flux” comes from the Latin participle *fluxus*, meaning flowing.

light: This term is reserved for visually evaluated radiant power. The process of visual evaluation is defined in **Section 4.1 Action Spectrum for Vision**. Light can be considered the luminous equivalent of power and is properly called luminous flux. *Light* is often used as shorthand for *luminous flux*, especially in applications. As is often the case, power is more easily and accurately measured than is energy, and this is the case with radiant quantities. In this practical sense, luminous power (light, or luminous flux) is more fundamental or basic than luminous energy (the time-quantity of light). It should be noted that this definition is entirely different from the use of this term in physics, where it is synonymous with radiant energy of any wavelength.

illumination: This term is reserved to describe the general circumstance of light incident on a surface or

body, or the general condition of being illuminated. It is used as a term of quality rather than quantity. The term of quantity is *illuminance* (see **Section 6.1 Illuminance**).

source: This is a general term used to reference a source of light. It can refer variously to an electric lamp, an LED, an entire luminaire with lamp and optical control, or fenestration for daylighting.

Finally, words such as *intensity* and *efficiency* are used in special and precise ways in lighting design and illuminating engineering, and their everyday meaning or the substitution of a seeming synonym can be misleading, if not incorrect. Thus, the term “intensity of illumination” is incorrect, and the term “visible light” is redundant.

2.2 Radiant and Luminous Concepts

Each concept involving radiation, light, and vision has a name, its quantification specified by a unit and its presence indicated by a symbol. In many cases a concept’s unit has a name. Concepts constructed from more fundamental ones have constituent units and names. In most cases a concise definition of a concept can be expressed as a mathematical equation using the symbols of the more fundamental concepts.

In general, words based on *radiate* refer to purely physical, radiant quantities, as in the case of *radiant power* and *optical radiation*. This is in distinction to *luminous*, which designates quantities involving radiant power that is visually evaluated. Some concepts have parallel radiant and luminous forms; one set used when optical radiation is considered simply as a physical entity, and another set when it is visually evaluated.

Radiometry and radiometric concepts deal with the measurement and conceptualization of radiant power as a physical entity; photometry and photometric concepts deal with visually evaluated radiant power. Photometric quantities always involve radiant power evaluated when the adaptation state is either photopic or scotopic. (Refer to ANSI/IES LS-7-20, *Lighting Science: Vision – Eye and Brain* for additional information.⁵⁾ If there are parallel radiometric and photometric concepts, the same symbol is used, with the radiometric symbol being augmented with a subscript, e.

2.3 Wavelength Dependencies

When it is necessary to indicate a quantity's dependence on the wavelength of the optical radiation involved, the adjective "spectral" is added to the name, and the standard parenthetical notation of mathematical functions is used, along with the universal symbol for wavelength, λ . As an example, Φ is the symbol for luminous flux, and $\Phi(\lambda)$ is the symbol for spectral luminous flux, that is, flux as a function of wavelength. When it is necessary to indicate how a quantity changes with wavelength, λ is used as a subscript to indicate the first derivative with respect to wavelength. Thus, the spectral luminous flux per unit wavelength is indicated by $\Phi_{\lambda}(\lambda)$ with $\Phi_{\lambda}(\lambda) = d\Phi(\lambda)/d\lambda$.⁴

3.0 Radiant Power, Radiant Flux

Electric light sources convert electrical power to radiant power, which is then emitted by the source. The emission can be conceived as either electromagnetic radiation or as photons. The following concepts and quantities are used to describe and quantify this power.

3.1 Specifying Radiant Energy and Power

3.1.1 Radiant Energy. This defines the electromagnetic or photonic radiant energy from a source.

Concept:	Energy emitted, transferred, or received in the form of radiation
Concept name:	radiant energy
Concept symbol:	$Q_e, Q_e(\lambda)$
Constituent units:	$\text{kg}\cdot\text{m}^2/\text{s}^2$
Unit name:	joule, J
Mathematical definition:	None

3.1.2 Radiant Power or Radiant Flux. This defines the electromagnetic or photonic radiant power from a source; that is, the time rate of flow of radiant energy.

Concept:	The rate of flow of electromagnetic or photonic radiation, the radiant power from a source.
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Concept name:	radiant flux
Concept symbol:	$\Phi_e, \Phi_e(\lambda)$
Constituent units:	joules per second, J/s
Unit name:	radiant watt
Mathematical definition:	$\Phi_e = \frac{dQ_e}{dt}$

3.1.3 Spectral Power Distribution. This expresses the radiant power emitted by a source of optical radiation over a range of particular wavelengths.

Concept:	Amount of optical radiation emitted by a source with wavelengths defined by a narrow band, $\Delta\lambda$, centered on a particular wavelength, λ
Concept name:	spectral power
Concept symbol:	$Q_e, Q_e(\lambda)$
Constituent units:	radiant watts per unit wavelength
Unit name:	None
Mathematical definition:	$\Phi_{e\lambda}(\lambda) = d\Phi_e(\lambda)/d\lambda$, with $\Phi_{e\lambda}(\lambda) \approx \Delta\Phi_e(\lambda)/\Delta\lambda$

3.1.4 Relative Spectral Power Distribution (SPD).

This is the quantity most commonly used in lighting to express the nature of radiant power emitted by a source. To make the spectral power distribution relative, all the data are divided by either the average value, the maximum value within the wavelength range of interest, or some arbitrarily chosen value. Although relative SPDs are provided in all practical work, the adjective "relative" is seldom used.^{4,6}

Concept:	Normalized spectral power distribution
Concept name:	relative spectral power
Concept symbol:	$S(\lambda)$
Constituent units:	relative radiant watts per unit wavelength
Unit name:	None
Mathematical definition:	$S(\lambda) = \Phi_{e\lambda}(\lambda)/R$, where R = some fixed value of $\Phi_{e\lambda}$

3.2 Data Conventions for SPDs

Some sources of optical radiation, such as incandescent sources, exhibit a continuous spectrum of radiant power over a wide range of wavelengths. Although the measurement process can only sample the spectrum at a discrete number of points, the data are usually presented as a continuum. **Figure 3-1** shows a continuous relative spectral power distribution of an incandescent lamp. The data are relative to the value at 750 nm, the wavelength at which the distribution is maximum in the visible region of the spectrum, expressed as a percentage of that maximum.

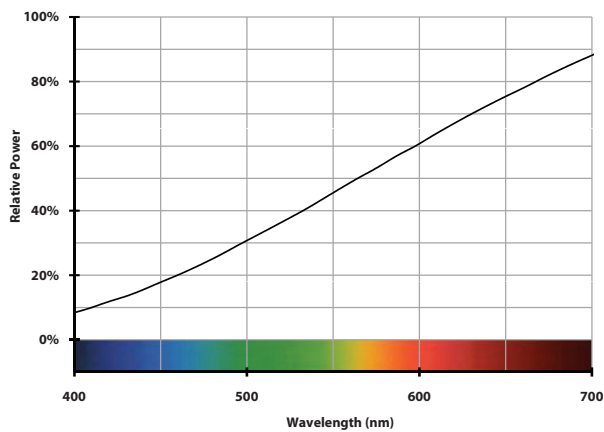


Figure 3-1. Relative spectral power distribution of a tungsten-halogen incandescent lamp operating at filament temperature of 3000 K. (© Illuminating Engineering Society)

Some sources of optical radiation emit radiant power only at a few discrete wavelengths or within very narrow ranges of wavelengths, each range centered on a particular wavelength. These are called line spectra. A low pressure mercury discharge is such a source.

4.0 Action Spectra

A photochemical effect produced by radiant power is said to be an actinic effect. Actinic effects can be direct, as in the case of chemical activity triggered by atoms or molecules absorbing photons, or indirect, as in the case of a high-level change in a biological organism produced by absorbed radiant power in photoreceptors. Actinic effects are usually the result of

complicated physical and chemical mechanisms that are affected by exposure time and previous exposure, and exhibit interactions (constructive or opponent) between wavelengths. But these mechanisms are usually ignored, and action spectra are used to simply link radiant input to the final actinic effect.⁷

Examples of actinic effects are the reaction of photodiodes (photoionization⁶), skin reddening (erythema), the bleaching of photopigments (isomerization⁴) in the rods and cones of the retina, and photosynthesis and phototropism in plants.

The action spectrum of an actinic effect is the magnitude of the effect produced by various wavelengths of monochromatic radiant power through some range of wavelengths. **Figure 4-1** shows the action spectrum for photosynthesis in green plants.

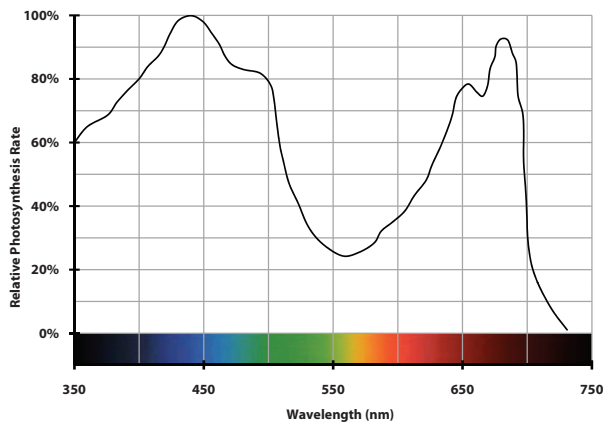


Figure 4-1. Photosynthesis action spectrum. The relative action spectrum of photosynthesis for common green plants. (© Illuminating Engineering Society)

Concept:	Photochemical effect of optical radiation of individual wavelengths over a range of wavelengths of interest
Concept name:	action spectrum
Concept symbol:	$\sigma(\lambda)$
Constituent units:	actinic response per unit wavelength

Unit name: None

Mathematical definition: $\sigma(\lambda) = R(\lambda)$,
or $\sigma(\lambda) = \text{Response}(\lambda)/R$,
where R = fixed value of
 $\text{Response}(\lambda)$

The units of an action spectrum depend on the actinic effect. In many cases, an action spectrum is normalized using its maximum value and so becomes a unitless efficiency function of wavelength.

By convention, the total actinic effect (TAE) of a source of optical radiation is defined by the wavelength-by-wavelength product of the spectral power distribution of the source and the action spectrum of the actinic effect:

$$\text{TAE} = K \int_{\lambda_1}^{\lambda_2} \sigma(\lambda) S(\lambda) d\lambda \approx K \sum_{i=1}^N \sigma(\lambda_i) S(\lambda_i) \Delta\lambda , \quad (4-1)$$

where:

λ_1 and λ_2 = limits of the wavelength range of interest

K = scaling constant for the action spectrum
and/or the spectral power distribution

It is important to understand that simply summing effects at individual wavelengths assumes that either the cumulative effect does not exhibit interactions between effects at different wavelengths, or that such interactions are negligible. In this case the process is said to be linearly additive. Strict linear additivity is rarely the case for real, total actinic responses, especially in biological effects. Nevertheless, linear additivity can be used to adequately represent the total response of some actinic effects for a wide spectrum of radiant power. Linear additivity implies both proportionality and that the total actinic effect of two sources is the sum of the two individual total effects:

$$\begin{aligned} \text{TAE} &= K \int_{\lambda_1}^{\lambda_2} \sigma(\lambda) [\alpha_1 S_1(\lambda) + \alpha_2 S_2(\lambda)] d\lambda \\ &= K \alpha_1 \int_{\lambda_1}^{\lambda_2} \sigma(\lambda) S_1(\lambda) d\lambda + K \alpha_2 \int_{\lambda_1}^{\lambda_2} \sigma(\lambda) S_2(\lambda) d\lambda \end{aligned} \quad (4-2)$$

5.0 Defining Light

The definition of light involves radiant power and the assessment of its efficacy using an action spectrum that must be, in some sense, a quantification of vision.

5.1 Action Spectrum for Vision

Light is defined as visually evaluated radiant power and it has been customary to use the process defined by **Equation 4-1** to perform this evaluation.⁹ This, in turn, requires that an indirect actinic effect be defined, presumably beginning with retinal photoreceptors changed by the absorption of optical radiation. This indirect actinic effect is then, in some sense, "vision," and the action spectrum then assigns to each wavelength a power to invoke "vision" or a visual sensation. It would be possible to define this sensation as any of the following: brightness, detection, recognition,¹⁰ conspicuity, or reaction time. The earliest attempt at such an assessment used recognition, but beginning with the work of Koenig,¹¹ brightness has been used to define the action spectrum of vision.

5.2 Photopic Luminous Efficiency

A photopic, brightness-based action spectrum was adopted internationally in 1924 by the CIE.¹² The data used to define this action spectrum resulted from a series of experiments that determined the relative brightness of monochromatic radiant power throughout the visible spectrum.^{1,13} The method involved comparing and equilibrating the brightness produced by radiant power at neighboring wavelengths, moving step-by-step through the spectrum. This avoided both the problem of matching brightness in the presence of large color differences and the use of flicker photometry. Foveal vision was used, with observers photopically adapted, using a 2-degree visual field. The inverse of the power required at each wavelength to produce a constant brightness is a measure of the efficacy of that wavelength. These data were made relative to the value at $\lambda = 555$ nm and thus defined a unitless efficiency function: the photopic luminous efficiency function of wavelength. Since the adoption of the standard values for this function, the CIE has modified and corrected them. Standard values given in 1983 are shown in **Table 5-1** and plotted in **Figure 5-1**.¹⁴ Recent research has proposed further modification.¹⁵

Concept:	Action spectrum of vision at photopic adaptation
Concept name:	photopic luminous efficiency function of wavelength

Concept symbol: $V(\lambda)$

Constituent units: None

Unit name: None

Mathematical definition: None

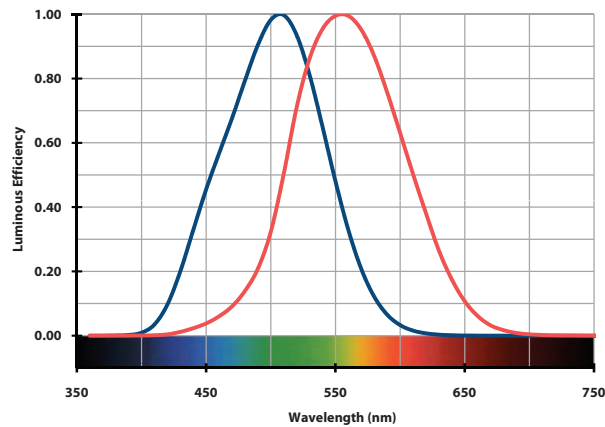


Figure 5-1. CIE Luminous efficiency functions of wavelength. The CIE 2° photopic and scotopic luminous efficiency functions of wavelength. The standard values are at 10-nm intervals, a smooth line is interpolated between them. (© Illuminating Engineering Society)

Table 5-1. CIE Standard 2° Photopic and Scotopic Luminous Efficiency Functions of Wavelength

Wavelength (nm)	$V(\lambda)$	$V'(\lambda)$	Wavelength (nm)	$V(\lambda)$	$V'(\lambda)$
360	0.0000	0.0000	570	0.9520	0.2076
370	0.0000	0.0000	580	0.8700	0.1212
380	0.0000	0.0006	590	0.7570	0.0655
390	0.0001	0.0022	600	0.6310	0.0332
400	0.0004	0.0093	610	0.5030	0.0159
410	0.0012	0.0348	620	0.3810	0.0074
420	0.0040	0.0966	630	0.2650	0.0033
430	0.0116	0.1998	640	0.1750	0.0015
440	0.0230	0.3281	650	0.1070	0.0007
450	0.0380	0.4550	660	0.0610	0.0003
460	0.0600	0.5670	670	0.0320	0.0001
470	0.0910	0.6760	680	0.0170	0.0001
480	0.1390	0.7930	690	0.0082	0.0000
490	0.2080	0.9040	700	0.0041	0.0000
500	0.3230	0.9820	710	0.0021	0.0000
510	0.5030	0.9970	720	0.0010	0.0000
520	0.7100	0.9350	730	0.0005	0.0000
530	0.8620	0.8110	740	0.0002	0.0000
540	0.9540	0.6500	750	0.0001	0.0000
550	0.9950	0.4810	760	0.0001	0.0000
560	0.9950	0.3288	770	0.0000	0.0000

5.3 Scotopic Luminous Efficiency

A scotopic, brightness-based action spectrum was adopted internationally in 1951 by the CIE.¹⁶ The data used to define this action spectrum resulted from experiments that determined the relative brightness of monochromatic optical radiation throughout the visible spectrum.^{4,17} A large, off-axis visual field of 20 degrees was used with observers scotopically adapted. The data were made relative to the value at $\lambda = 505$ nm and thus defined a unitless efficiency function. Standard values at 10-nm intervals given in 1983 are shown in **Table 5-1** and plotted in **Figure 5-1** (both in **Section 5.2**).

Concept: Action spectrum of vision at scotopic adaptation

Concept name: scotopic luminous efficiency function of wavelength

Concept symbol: $V'(\lambda)$

Constituent units: None

Unit name: None

Mathematical definition: None

6.0 Luminous Flux

Luminous flux is visually evaluated radiant flux and defines “light” for purposes of lighting design and illuminating engineering. Following the customary use of action spectra, radiant flux is evaluated wavelength-by-wavelength using either of the two standard action spectra for vision: the photopic or scotopic luminous efficiency functions of wavelength. The sum of the individual wavelength evaluations defines the total effect.

6.1 Photopic Luminous Flux

This is the most common unit of light. It can be considered photopic luminous power and, akin to radiant power, is the time rate of flow of the quantity of photopic light. The constant 683 scales the total visually evaluated radiant watts of the source to the modern photometric unit of the photopic lumen.

Concept:	The flow of photopic luminous power from a source
Concept name:	photopic luminous flux
Concept symbol:	Φ
Constituent units:	None
Unit name:	photopic lumen, lm
Mathematical definition:	
$\Phi_V \equiv 683 \int_0^\infty \Phi_{e\lambda}(\lambda) V(\lambda) d\lambda \approx 683 \sum_{\lambda=360}^{830} \Phi_{e\lambda}(\lambda) V(\lambda) \Delta\lambda$	

Figure 6-1 illustrates the light output or luminous flux leaving the lamp in nearly all directions.

The most common analogy is fluid flux, which is the time rate of flow of water, as shown in **Figure 6-2**; when the garden hose valve is opened, streams of water emerge and travel outward.

A second analogy is the electrical one, in which luminous flux is compared to electric current, in amperes, which is the rate of flow of electric charge through a wire.

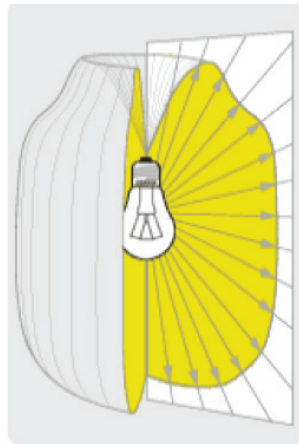


Figure 6-1. Light output illustration. (© Illuminating Engineering Society)



Figure 6-2. Garden hose. (© iStock/ChepeNicolli)

6.2 Scotopic Luminous Flux

An uncommon unit of light, it can be thought of as scotopic luminous power. The constant 1,700 scales the total visually

evaluated radiant watts of the source to the modern photometric unit of the scotopic lumen and results from the assumption that when using the $V'(\lambda)$ function, its values are all scaled so that $V'(555 \text{ nm}) = V(555 \text{ nm})$.

Concept:	The flow of scotopic luminous power from a source
Concept name:	scotopic luminous flux
Concept symbol:	$\Phi_{V'}$
Constituent units:	None
Unit name:	scotopic lumen
Mathematical definition:	
$\Phi_{V'} \equiv 1,700 \int_0^\infty \Phi_{e\lambda}(\lambda) V'(\lambda) d\lambda \approx 1,700 \sum_{\lambda=360}^{830} \Phi_{e\lambda}(\lambda) V'(\lambda) \Delta\lambda$	

6.3 Quantity of Light

This is luminous power integrated over time; the luminous equivalent of energy. The quantity of light may arise when total light exposure is of interest, as happens when dealing with plants, or assessing the possible damage light might cause to a piece of art, or when medical light dosage is to be considered.

Concept:	The time-integrated amount of light
Concept name:	quantity of light
Concept symbol:	Q_V
Constituent units:	lumens, seconds
Unit name:	lumen-seconds
Mathematical definition:	$Q_V = \int \Phi_V dt$

6.4 Luminous Efficacy of Radiation

This efficacy is reserved to describe a characteristic of radiation: the ratio of the lumens it contains to its power in watts. Though uncommon when referring to electric light sources, efficacy of radiation is used to describe the optical radiation from the sun and sky in daylighting applications.

Concept:	The ratio of luminous power to radiant power
Concept name:	luminous efficacy of radiation
Concept symbol:	K
Constituent units:	lumens, radiant watts
Unit name:	None
Mathematical definition:	$K = \Phi / \Phi_e$

6.5 Luminous Efficacy of a Source

This efficacy is reserved to describe a characteristic of a source of radiation: the ratio of the lumens emitted to the watts required to produce the radiation that contains those lumens. This efficacy is a frequently cited characteristic of electric light sources and provides a measure of how effectively they convert electric power to luminous power.

Concept:	The ratio of luminous power to the power consumed by the source
Concept name:	luminous efficacy of a source
Concept symbol:	η , lm/W, LPW
Constituent units:	lumens, watts
Unit name:	None
Mathematical definition:	$\eta \equiv \Phi/W$

7.0 Surface Flux Densities

The most common concepts used to quantify aspects of lighting involve not the absolute amount of luminous flux but rather the density of flux. Quantities involving flux density onto or from a surface are used in lighting to state some design recommendations and to describe the final luminous condition of a task or architectural surface.

7.1 Illuminance

Illuminance is the incident luminous flux density on a differential element of surface located at a point and oriented in a particular direction, expressed in lumens per unit area. Since the area involved is differential, it is customary to refer to this as *illuminance at a point*. The unit name depends on the constituent unit for area. It is lux if square meters are used for area, and footcandles if square feet are used.

Concept:	Local surface density of incident luminous flux
Concept name:	illuminance
Concept symbol:	E
Constituent units:	lumens, area
Unit name:	lux (lumens/square meter), lx footcandle (lumens/square foot), fc
Mathematical definition:	$E \equiv d\Phi_{\text{on}}/dA$

Aside from the general notion that flux is incident, illuminance does not describe the amount arriving from various directions, only the total incident. Without additional information, this can limit the utility and significance of illuminance. **Figure 7-1** shows two very different illumination conditions that have the same illuminance. On the left, all the flux arrives at the surface from the same direction; on the right it arrives uniformly from all directions. In both cases the density of lumens to area is the same.

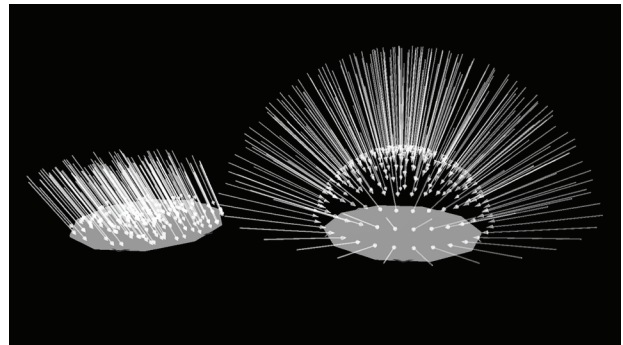


Figure 7-1. Two different illumination conditions that have the same illuminance. (© Illuminating Engineering Society)

7.1.1 Average Illuminance. In certain circumstances knowing the average illuminance over a large area is useful in the lighting design or analysis process. Like any simple average, average illuminance reveals nothing about any local variations in illuminance that might exist over the area for which it is determined; nevertheless, it can describe in a general way a useful attribute of a lighted surface.

Concept:	Mean surface density of incident luminous flux over an extended area
Concept name:	average illuminance
Concept symbol:	$\bar{E}$, E_{avg}
Constituent units:	lumens, area
Unit name:	lux (lumens/square meter), lx footcandle (lumens/square foot), fc

Mathematical definition:

$$\bar{E} \equiv \Phi_{\text{on}}/A = \frac{1}{A} \int_0^A E dA \approx \frac{1}{A} \sum_{i=1}^N \Delta A_i E_i = \frac{\Delta A}{A} \sum_{i=1}^N E_i = \frac{1}{N} \sum_{i=1}^N E_i$$

7.2 Exitance

Exitance is the exitant (leaving) luminous flux density

on a differential element of surface located at a point, expressed in lumens per unit area. Exitance is emitted (reflected or transmitted) flux density, and so can be related to how luminous the emitting surface is or how bright it appears. Exitance does not have a named unit; “lumens per square meter” or “lumens per square foot” is used when describing exitance.

Concept:	Local surface density of emitted luminous flux
Concept name:	exitance
Concept symbol:	M
Constituent units:	lumens, area
Unit name:	None
Mathematical definition:	$M \equiv d\Phi_{\text{off}}/dA$

Like illuminance, exitance does not provide information about the directions into which the surface emits flux, only the total amount. **Figure 7-2** shows extreme cases of two surfaces with identical exitance but radically different emitting characteristics. On the left, all the flux leaves the surface into the same direction; on the right it leaves uniformly into all directions. In both cases the density of exitant lumens to area is the same.

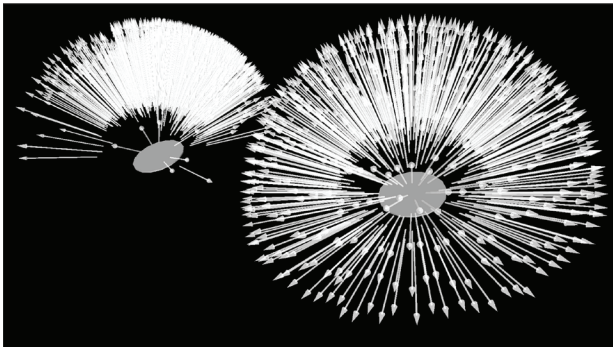


Figure 7-2. Two different emitting conditions that have the same exitance. (© Illuminating Engineering Society)

Exitance is useful in that it describes the general light emitting power of a surface. However, because of its non-directionality, it may not indicate how luminous an object or surface appears from a particular point of view. Only in the case of a surface emitting flux diffusely can a reliable relationship be established between exitance and luminance. (See **Section 8.3 Luminance**.)

7.2.1 Average Exitance. As with average illuminance, knowing the average exitance over a large area is useful in the lighting design or analysis process, but it reveals

nothing about any local variations that might exist over the area for which it is determined.

Concept:	Mean surface density of emitted luminous flux
Concept name:	average exitance
Concept symbol:	$\bar{M}$, M_{avg}
Constituent units:	lumens, area
Unit name:	None

Mathematical definition:

$$\bar{M} \equiv \Phi_{\text{off}}/A = \frac{1}{A} \int_0^A M dA \approx \frac{1}{A} \sum_{i=1}^N \Delta A_i M_i = \frac{\Delta A}{A} \sum_{i=1}^N M_i \approx \frac{1}{N} \sum_{i=1}^N M_i$$

or

$\bar{M} = \bar{E}\rho$ for a reflecting surface, where ρ is the reflectance of the surface

or

$\bar{M} = \bar{E}\tau$ for a transmitting surface, where τ is the transmittance of the material

8.0 Spatial Flux Densities

In order to describe the density of flux in space, a measure of “space” is required. This is not volume but rather a quantity that describes the apparent extent or size of an object from a point of regard.

8.1 Solid Angle

Solid angle is used to define spatial extent for the purposes of establishing spatial flux densities. Just as *plane angle* specifies the extent of separation between two intersecting lines of indeterminate length, *solid angle* specifies the extent of a cone of indeterminate length.

The solid angle, denoted by the Greek letter “omega” (ω), is defined as the ratio of the intercepted surface area of a sphere centered on that point to the square of the sphere’s radius, thus:

$$\omega = \frac{A}{r^2} \quad (8-1)$$

Figure 8-1 shows this relationship graphically, where r is the radius of the sphere, and A is the area of the surface subtended by the solid angle.

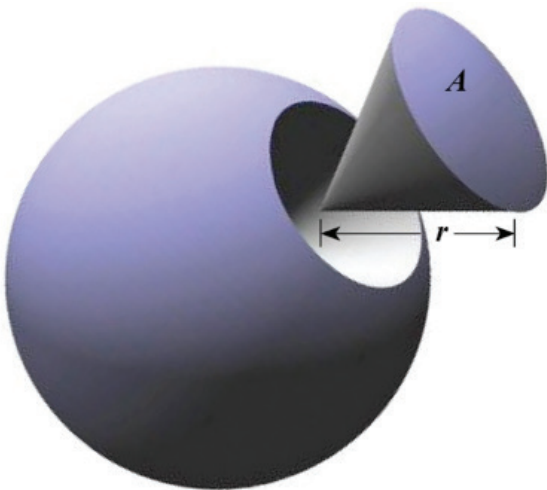


Figure 8-1. Solid angle. (© Illuminating Engineering Society)

Figure 8-2 shows another cone of solid angle and how three discs of different sizes and orientations can exhibit the same solid angle from a point of regard. Though of different surface extent and orientation, they have the same spatial extent with respect to the apex of the cone, the point of regard. Solid angles are measured in steradians (sr).

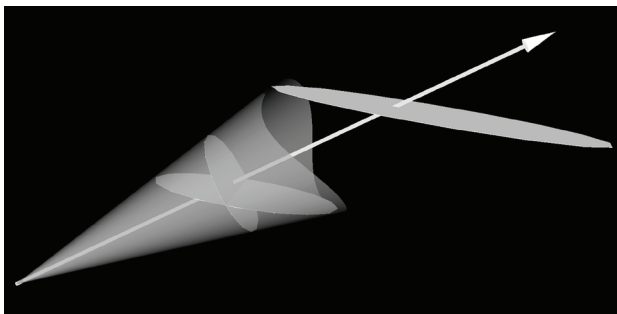


Figure 8-2. The solid angle (represented by the open-ended cone) for three discs of different sizes and orientations. (© Illuminating Engineering Society)

Concept:	Spatial extent
Concept name:	solid angle
Concept symbol:	ω
Constituent units:	area, distance
Unit name:	steradian, sr
Mathematical definition:	$d\omega \equiv dA \cos \theta / D^2$; $\omega = \int_A dA \cos \theta / D^2$, where in this case D is the distance and θ is the angle of incidence to the surface.

8.2 Luminous Intensity

Luminous intensity specifies the light emitting power of a point source in a particular direction and is defined as the density of luminous flux in space in that direction, or the number of lumens per steradian. Luminous intensity is also called *candlepower*. It is common to use the spherical coordinate system to specify a direction from a point source, and so the luminous intensity distribution of a source is often expressed as a function of the two spherical coordinate angles. Luminous intensity is invariant with distance from the source. **Figures 8-3** and **8-4** show how luminous intensity describes the spatial distribution of light from a source.

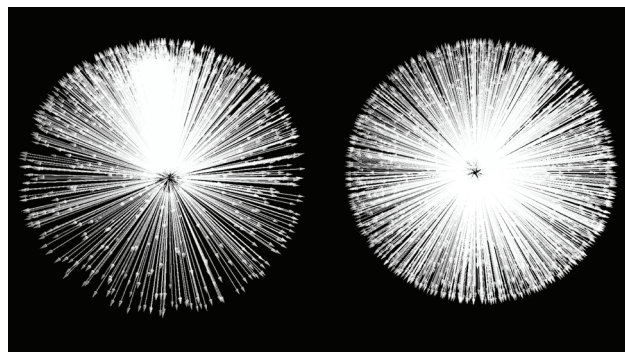


Figure 8-3. Spatial distribution of flux for two sources, indicated by the density of rays emitted in various directions. The source on the left emits more light in the upward direction, while that the one on the right distributes light more or less uniformly in all directions. (© Illuminating Engineering Society)

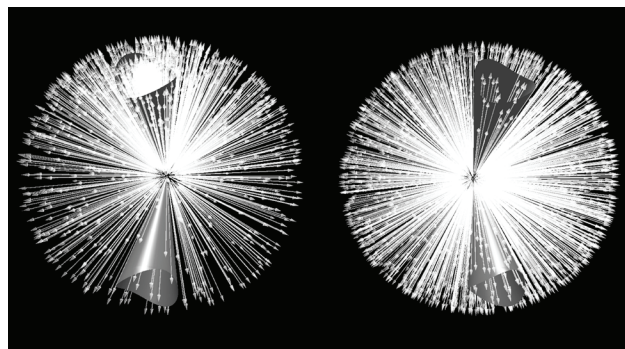


Figure 8-4. Luminous intensities for two sources. For each source, two cones of solid angle are positioned with their apexes at the source. The number of rays within each cone is a measure of the density, in lumens per steradian, that the source established, and thus its luminous intensity in that direction. (© Illuminating Engineering Society)

Concept:	Spatial density of luminous flux from a point source
Concept name:	luminous intensity (candlepower)
Concept symbol:	I
Constituent units:	lumens, steradians
Unit name:	candela, cd, lm/sr
Mathematical definition:	$I(\theta, \psi) \equiv d\Phi(\theta, \psi)/d\omega$ (see Figure 8-5)

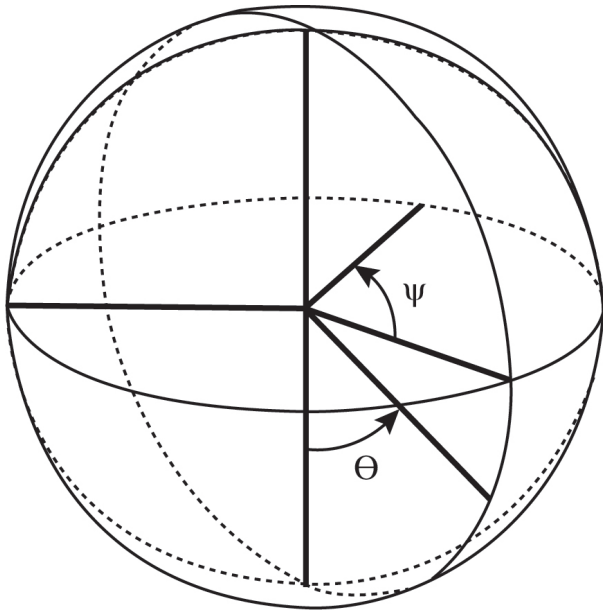


Figure 8-5. Showing vertical angle θ and azimuthal angle ψ . (© Illuminating Engineering Society)

Luminous intensity is analogous to pressure in a fluid system and voltage in an electric system.

Luminous flux (Φ) and luminous intensity (I) are directly related through solid angle:

$$\Phi = I\omega, \text{ or } I = \frac{\Phi}{\omega} \quad (8-2)$$

8.2.1 Equivalent Luminous Intensity. An operational definition of luminous intensity can be used to approximately describe the light emitting power of sources that are luminous areas rather than points. The illuminance E produced by a point source at a point on a surface located a distance D from the source and oriented so that the surface perpendicular points directly back to the source is:

$$E = \frac{I(\theta, \psi)\cos\theta}{D^2} = \frac{I(\theta, \psi)\cos 0^\circ}{D^2} = \frac{I(\theta, \psi)}{D^2}, \quad (8-3)$$

where:

$I(\theta, \psi)$ = luminous intensity of the point source in the direction of the illuminated point

D = distance from point source to the illuminated point

Equation 8-3 is inverted to give an operational definition of *luminous intensity*:

$$\bar{I}(\theta, \psi) = ED^2 \quad (8-4)$$

That is, intensity can be operationally defined as the product of the illuminance it produces at some distant point and the square of the distance to that point.

If an area source produces an illuminance, E' , at a point some distance D from its center and in a particular direction (θ, ψ) , then **Equation 8-4** gives an operational definition of luminous intensity of this area source. This is the equivalent luminous intensity, $\bar{I}$, of the area source. It should be noted that equivalent luminous intensity is not invariant with distance, since for a real area source the ratio of illuminance produced to distance-squared does not remain constant with distance. In practice, relatively large distances are used and equivalent luminous intensity is the quantity used to describe the distribution of light from virtually all practical lighting equipment. This photometric procedure is described in detail in *ANSI/IES LS-4-20(R2023), Lighting Science: Measurement of Light – The Science of Photometry*.⁶

8.3 Luminance

Luminance is a measure of the light emitting power of a surface in a particular direction per unit apparent area, sometimes referred to as *projected area*. This is expressed as the density of luminous intensity per unit apparent area. Implicit in the definition is the assumption that the area is small.

Luminance is perhaps the most important quantity in lighting design and illuminating engineering, as it is one of the direct stimuli to vision, and many measures of performance and perception have been shown to depend on luminance. **Figure 8-6** depicts the definition of luminance. The direction in which the luminance is determined is indicated by the dark arrow, and the

angle of view, θ , is measured from this direction to the surface perpendicular.

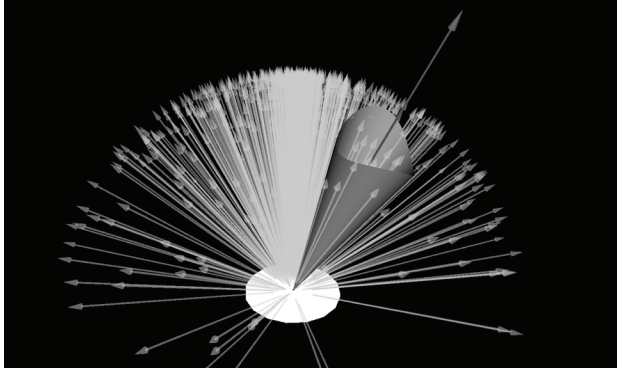


Figure 8-6. The luminance of a surface is the luminous intensity (lumens per steradian) in a particular direction per unit apparent area. The light distribution of a surface may be non-uniform (as shown here). (© Illuminating Engineering Society)

Concept:	Local surface density of light emitting power in a particular direction.
Concept name:	luminance
Concept symbol:	$L(\theta, \psi)$
Constituent units:	luminous intensity, area
Unit name:	candela per square meter (nit)

Mathematical definition:

$$L(\theta, \psi) \equiv \frac{dI(\theta, \psi)}{dA \cos \theta} = \frac{d^2 \Phi}{d\omega dA \cos \theta}$$

The mathematical definition also establishes an *operational definition*: the luminance of a surface is the ratio of the illuminance it produces at a distant point, to the solid angle it subtends at that point.¹⁸

$$L \equiv \frac{dE}{d\omega} \cos \theta \quad (8-5)$$

Equation 8-3 expresses this operational definition and is the basis for all luminance meters: an illuminance measurement made through a cone of known solid angle. **Equation 8-3** also shows that a surface need not be involved to establish a luminance.

Average luminance ($\bar{L}$, L_{avg}) can be defined and approximated for a large area:

$$\bar{L} = \int_A \frac{dI(\theta, \psi)}{dA \cos \theta} \approx \frac{\bar{I}(\theta_{\text{avg}}, \psi_{\text{avg}})}{A \cos \theta_{\text{avg}}} \quad (8-6)$$

9.0 Light and Materials

The interaction of the light and materials is an important aspect of architectural lighting. The following concepts are used to define these interactions, involving not only the quantity of lighting but the types of spatial distributions that result.

9.1 Reflectance

Reflectance is the ratio of exitant (reflected) to incident luminous flux. It may or may not be specified with regard to the incident or exitant directions. Reflectance may involve the sum of all luminous wavelengths or be determined as a function of wavelength, in which case it is *spectral reflectance*.

Reflectance is affected by the geometry, wavelength, and polarization of the incident flux.⁴

Concept:	The fraction of incident light that is reflected by a surface
----------	---

Concept name: reflectance

Concept symbol: ρ , $\rho(\lambda)$

Constituent units: lumens

Unit name: None

Mathematical definition: $\rho \equiv \Phi_{\text{off}} / \Phi_{\text{on}} ; 0 \leq \rho \leq 1$

One common system for specifying the geometry of incident and reflected flux uses cones and hemispheres to define the extent and direction of flux. Incident flux can be specified as arriving from a particular direction in a cone, or uniformly from all directions in a hemisphere. Similarly, reflected flux can be specified as exitant in a particular direction in a cone, or into any direction within a hemisphere. The cones involved can be small but finite or vanishingly small, in which case a single direction is involved. In all cases, the limiting values are 0 and 1 since reflectance is defined as the ratio of luminous fluxes. In special cases involving fluorescent materials, it is possible to have a reflectance greater than 1 if the photopic (or scotopic) response of the light emitted by the fluorescent material is greater than the photopic response of the light absorbed by the phosphor.

The most common arrangement used to measure and specify reflectance for architectural surfaces is conical-incident and hemispherical-exitant. Since the geometry is fixed, a single value defines the reflective power of the surface. As described in *ANSI/IES LS-4-20(R2023), Lighting Science: Physics and Optics of Radiant Power*,⁴ reflectances can be specular, diffuse, or spread. **Figure 9-1** depicts diffuse and specular reflectance. Diffuse reflectance (left) sends light uniformly in all directions regardless of the incident direction. Specular reflectance (right) sends light into the plane formed by the incident ray and the surface perpendicular, and at an angle from that perpendicular equal to that of the incident ray. Thus, in specular reflectance the light is conical-exitant, and the exitant cone is the same size as the incident cone.

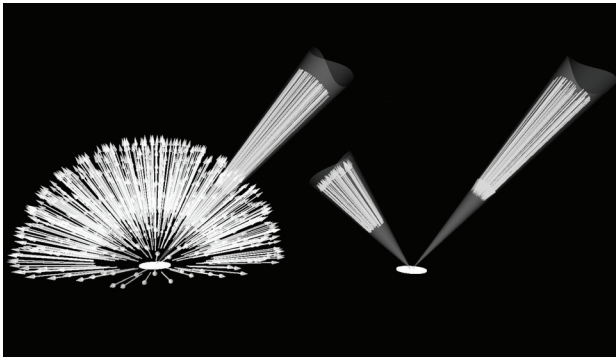


Figure 9-1. Diffuse (left) and specular (right) reflectance.
(© Illuminating Engineering Society)

9.1.1 Perfectly Diffuse Reflectance: A Useful Special Case. Most practical illuminated surfaces reflect incident light into many directions. This property can be extended to define a hypothetical surface that exhibits a distribution of reflected light such that its density varies with the cosine of the exitant angle measured from the surface perpendicular. This special reflected distribution is called perfectly diffuse reflectance. It is important to note that perfect diffuseness does not mean a uniform distribution, but rather a distribution that is most dense in the direction of the surface perpendicular, decreasing as the cosine of the angle of the reflected direction. Note also that perfect diffusion does not mean perfect reflection; that is, it does not mean a reflectance of 1.0.

Surfaces that are perfectly diffuse reflectors, also called *Lambertian surfaces*, exhibit this distribution regardless

of the incident direction of light. One consequence of diffusely reflected light is that such a surface exhibits a luminance that is constant and independent of view. Another is that very great simplification of lighting calculations is possible.¹⁸ In the absence of more detailed information about illuminated surfaces, it is universally assumed within the lighting design process that surfaces are perfectly diffuse reflectors.

Concept:	Perfect diffusion of incident light by scattering and reflection
Concept name:	perfectly diffuse reflectance
Concept symbol:	$\rho, \rho(\lambda)$
Constituent units:	lumens
Unit name:	None
Mathematical definition:	$\rho \equiv \Phi_{\text{off}}(\text{diffuse})/\Phi_{\text{on}} ; 0 \leq \rho \leq 1$

9.1.2 Bidirectional Reflectance. In some cases, the form, texture, composition, or structure of a surface gives it reflectances that are strongly directional, and a single value is cannot adequately describe the surface's interaction with light. In these cases incident and exitant directions need to be accounted for, and multiple values of reflectance are necessary to characterize the surface.

The conceptually simplest bidirectional reflectance assumes conical-incident conical-exitant geometry and the reflectance is a function of the two directions. It is common to use the spherical coordinate system to specify these directions and so the bidirectional reflectance is the ratio of the luminous fluxes in the incident and exitant cones:

$$\rho(\theta_i, \psi_i; \theta_r, \psi_r) = \Phi(\theta_r, \psi_r) / \Phi(\theta_i, \psi_i); 0 \leq \rho \leq 1 \quad (9-1)$$

where:

(θ_i, ψ_i) = incident direction

(θ_r, ψ_r) = exitant (reflected) direction

9.1.3 Bidirectional Reflectance Distribution Function.

An alternative and more common way to specify directional reflectance is the bidirectional reflectance distribution function (BRDF), f_r . It has the advantage of being simpler to measure in practice than directional conical-conical reflectance. BRDF is defined as:

$$f_r(\theta_i, \psi_i; \theta_r, \psi_r) = dL_r(\theta_r, \psi_r) / E_i(\theta_i, \psi_i); 0 \leq f_r \leq \infty, \quad (9-2)$$

where:

$E_i(\theta_i, \psi_i)$ = illuminance produced by flux from the incident direction (θ_i, ψ_i)

$L_r(\theta_r, \psi_r)$ = luminance of the surface in the exitant (reflected) direction (θ_r, ψ_r)

The units of f_r are inverse steradians, sr^{-1} . BRDF's have been used to characterize visual tasks that do not exhibit perfectly specular or diffuse reflection for purposes of predicting visual performance,¹⁹ and to characterize the detailed reflecting properties of illuminated surfaces for computer graphic rendering of illumination and lighting systems.²⁰

9.2 Transmittance

Transmittance is the ratio of emergent to incident luminous flux. It may or may not be specified with regard to the incident or emergent (transmitted) directions. Transmittance may involve the sum of all luminous wavelengths or be determined as a function of wavelength, in which case it is *spectral transmittance*. The cone-hemisphere system of geometry used for reflectance is also used for transmittance. Limiting values are 0 and 1 since transmittance is the ratio of luminous fluxes.

Transmittance is affected by the geometry, wavelength, and polarization of the incident flux.⁴

Concept:	The fraction of incident light that passes through and exits a material.
Concept name:	transmittance
Concept symbol:	$\tau, (\tau)\lambda$
Constituent units:	lumens
Unit name:	None
Mathematical definition:	$\tau \equiv \Phi_{\text{out}} / \Phi_{\text{in}} \quad 0 \leq \tau \leq 1$

Figure 9-2 shows the two kinds of transmittance common in illuminated materials: diffuse and image-preserving. Diffuse transmittance (left) sends light uniformly in all directions regardless of the incident direction. Image preserving transmittance (right) preserves the direction in which the light travels. As a practical matter, there is always refraction, which offsets the rays, even in thin media with parallel faces.⁴

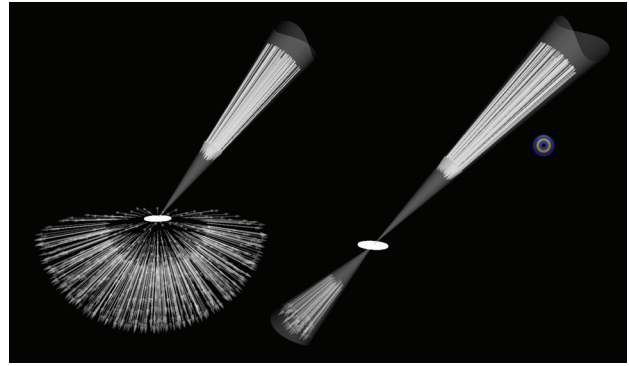


Figure 9-2. Diffuse and image preserving transmittance.
(© Illuminating Engineering Society)

9.2.1 Perfectly Diffuse Transmittance: A Useful Special Case. Some practical architectural materials redirect transmitted incident light into many directions. This property can be extended to define a hypothetical surface that exhibits a distribution of transmitted light such that its density varies with the cosine of the exitant angle measured from the surface perpendicular. This special transmitted distribution is called perfectly diffuse transmittance, and the surface is sometimes called a *Lambertian surface*. It is important to note that perfect diffuseness does not mean a uniform distribution, but rather a distribution that is most dense in the direction of the surface perpendicular, decreasing as the cosine of the angle of the transmitted direction. It should also be noted that perfect transmittance does not mean perfect transmittance; that is, it does not mean a transmittance of 1.0.

9.2.2 Bidirectional Transmittance. In some cases, the form, texture, composition, or structure of a surface give it transmittances that are strongly directional, and a single value is cannot adequately describe the surface's interaction with light. In these cases incident and exitant directions must be accounted for, and multiple values of transmittance are necessary to characterize the surface.

The conceptually simplest bidirectional transmittance assumes conical-incident conical-exitant geometry, and the transmittance is a function of the two directions. It is common to use the spherical coordinate system to specify these directions, and so the bidirectional transmittance is the ratio of the luminous fluxes in the incident and exitant cones:

$$\tau(\theta_i, \psi_i; \theta_t, \psi_t) = \Phi(\theta_i, \psi_i) / \Phi(\theta_t, \psi_t) ; 0 \leq \tau \leq 1 \quad (9-3)$$

where:

(θ_i, ψ_i) = incident direction

(θ_t, ψ_t) = exitant (transmitted) direction

9.2.3 Bidirectional Transmittance Distribution

Function. An alternative, and more common for specifying directional reflectance, is the bidirectional transmittance distribution function (BTDF), f_t . It has the advantage of being simpler to measure in practice than directional conical-conical transmittance. BTDF is defined as:

$$f_t(\theta_i, \psi_i; \theta_t, \psi_t) = dL_t(\theta_t, \psi_t) / E_i(\theta_i, \psi_i) ; 0 \leq f_t \leq \infty, \quad (9-4)$$

where:

$E_i(\theta_i, \psi_i)$ = illuminance produced by flux from the incident direction (θ_i, ψ_i)

$L_t(\theta_t, \psi_t)$ = luminance of the surface in the exitant (transmitted) direction (θ_t, ψ_t)

9.3 Absorptance

Absorptance defines the luminous flux that is absorbed by a material as flux passes through it. Whatever flux is not reflected or transmitted is absorbed.

Concept: The fraction of incident light that is lost in the interior of a material

Concept name: absorptance

Concept symbol: $Q_e, Q_e(\lambda)$

Constituent units: lumens

Unit name: None

Mathematical definition:

$$\alpha \equiv \frac{\Phi_{\text{lost}}}{\Phi_{\text{on}}} = \frac{\Phi_{\text{on}} - \Phi_{\text{out}}}{\Phi_{\text{on}}} ; 0 \leq \alpha \leq 1$$

10.0 Other Derived Concepts

Concepts derived from simpler ones are often used in lighting. Examples are contrast, used to specify one characteristic of a visual task, and brightness, the perceptual response to luminance.

10.1 Luminance Contrast

This term specifies the luminance difference exhibited by a visual target or object of interest from its immediate surround or background. Examples of visual target and background are the print on this page and the paper immediately around it. Luminance contrast can be negative, as is the case for dark print on white paper: the target luminance (luminance of the printed letters) is less than the background luminance (luminance of the paper). Sometimes *contrast* is defined absolutely; that is, it is always positive. In some cases, *contrast* is defined as a modulation that involves both the difference in the luminances and their summation.²¹

Concept: The luminance difference between a visual target and its immediate surround, relative to the surround

Concept name: luminance contrast

Concept symbol: C

Constituent units: luminance

Unit name: None

Mathematical definitions:

$$C = \frac{L_t - L_b}{L_b}, \text{ or } C = \left| \frac{L_t - L_b}{L_b} \right|, \text{ or } C = \frac{L_t - L_b}{L_t + L_b}$$

10.2 Brightness

Brightness is the perceptual response to luminance, is associated with the luminous power of a surface or object, and ranges from bright to dim. It is affected by luminance, surround luminance, adaptation, gradient, and spectrum.²¹

Concept: The strength of the luminous sensation from a visual stimulus; the visual response counterpart to luminance

Concept name: brightness

Concept symbol: B

Constituent units: None

Unit name: None

Mathematical definition: $B \propto L_t^{1/3} - B_0(L_b, \alpha)$

11.0 Tabulation

11.1 Radiometric Units

Some of the concepts shown in **Table 11-1** are listed without having been explained previously. They are the radiant or quantum (photon) equivalent of a similarly named photometric unit, and their significance should be clear.

11.2 Principal Photometric Units

Table 11-2 summarizes the principal photometric units commonly used in lighting. In each case the concept and concept name are provided. In some cases the concept unit has no name, as in the case of exitance. In other cases, the official name is seldom used and the constituent units are more common, as in the case of luminance, where the unit name is *nit* but the more common practice is to use cd/m^2 . In all cases, the mathematical equations express the definition of the quantity and are not necessarily used in practical computation.¹⁸

Table 11-1. Radiometric Quantities

Concept	Concept Name	Constituent Units	Symbol	Unit Name	Formula
Radiant energy	energy		Q_e	joule	
Radiant flux	power	energy, time	F_e	watt	$\Phi_e = \frac{dQ_e}{dt}$
Spectral power	power per unit wavelength	watt, length	$P(\lambda)$		$P(\lambda) = \frac{\Phi_e(\lambda)}{\Delta\lambda}$
Incident surface power density	irradiance	watt, area	E_e		$E_e = \frac{d\Phi_{e,\text{on}}}{dA}$
Exitant surface power density	radiant exitance	watt, area	M_e		$M_e = \frac{d\Phi_{e,\text{off}}}{d\omega}$
Spatial radiant power density	radiant intensity	watt, steradian	I_e		$I_e(\theta, \psi) = \frac{d\Phi_e(\theta, \psi)}{d\omega}$
Radiant intensity per unit area	radiance	radiant intensity, area	L_e		$L_e(\theta, \psi) = \frac{dI_e(\theta, \psi)}{dA \cos\theta} = \frac{d^2\Phi_e}{d\omega dA \cos\theta}$

Table 11-2. Photometric Quantities

Concept	Concept Name	Constituent Units	Symbol	Unit Name	Formula
Photopic visually evaluated radiant power	photopic luminous flux	lightwatts, lumens per watt	Φ	lumen, lm	$\Phi \equiv 683 \int_0^\infty \Phi_{e\lambda}(\lambda) V(\lambda) d\lambda$
Scotopic visually evaluated radiant power	scotopic luminous flux	lightwatts, lumens per watt	Φ'	lumen, lm	$\Phi \equiv 1,700 \int_0^\infty \Phi_{e\lambda}(\lambda) V'(\lambda) d\lambda$
Time-integrated amount of luminous flux, dosage	quantity of light	lumen	$Q_v, \Phi \cdot s$	lumen-second	$Q_v = \int \Phi dt$
Efficacy of radiation	efficacy	lumens, radiant watts	K		$K = \frac{\Phi}{\Phi_e}$
Efficacy of a source	efficacy	lumens, electrical watts	η		$\eta \equiv \frac{\Phi}{W}$
Incident surface flux density	illuminance	lumens, area	E	lux (lx), foot-candle (fc)	$E \equiv \frac{d\Phi_{on}}{dA}$
Emergent surface flux density	exitance	lumens, area	M		$M \equiv \frac{d\Phi_{off}}{dA}$
Spatial extent	solid angle	area, distance	ω	steradian, sr	$d\omega \equiv \frac{dA \cos\theta}{D^2};$ $\omega = \int_A \frac{dA \cos\theta}{D^2}$
Spatial flux density	luminous intensity	lumens, steradians	I	candela, cd	$I(\theta, \psi) \equiv \frac{d\Phi(\theta, \psi)}{d\omega}$
Spatial flux density emitted by a surface	luminance	candelas, area	L	cd/m ²	$L(\theta, \psi) \equiv \frac{dI(\theta, \psi)}{dA \cos\theta}$ $= \frac{d^2\Phi}{d\omega dA \cos\theta} \approx \frac{I(\theta, \psi)}{A \cos\theta}$
Fraction of incident optical radiation reflected by a material	reflectance	lumens	ρ		$\rho \equiv \frac{\Phi_{off}}{\Phi_{on}}; 0 \leq \rho \leq 1$
Reflectance of optical radiation as a function of wavelength	spectral reflectance	lumens	$\rho(\lambda)$		$\rho(\lambda) \equiv \frac{\Phi(\lambda)_{off}}{\Phi(\lambda)_{on}}; 0 \leq \rho \leq 1$
Reflectance of optical radiation from one direction into another	bidirectional reflectance	lumens	$\rho(\theta_i, \psi_i; \theta_r, \psi_r)$		$\rho(\theta_i, \psi_i; \theta_r, \psi_r) = \frac{\Phi(\theta_r, \psi_r)}{\Phi(\theta_i, \psi_i)}; 0 \leq \rho \leq 1$

Table 11-2. Photometric Quantities (continued)

Concept	Concept Name	Constituent Units	Symbol	Unit Name	Formula
Reflected luminance per unit illuminance of a surface	bidirectional reflectance distribution function	luminance illuminance	$f_r(\theta_i, \psi_i; \theta_r, \psi_r)$	sr^{-1}	$f_r(\theta_i, \psi_i; \theta_r, \psi_r) = \frac{dL_r(\theta_r, \psi_r)}{E_i(\theta_i, \psi_i)}; 0 \leq f_r < \infty$
Fraction of incident light through a material	transmittance	lumens	τ		$\tau \equiv \frac{\Phi_{\text{out}}}{\Phi_{\text{on}}}; 0 \leq \tau \leq 1$
Transmittance of optical radiation as a function of wavelength	spectral transmittance	lumens	$\tau(\lambda)$		$\tau(\lambda) \equiv \frac{\Phi(\lambda)_{\text{out}}}{\Phi(\lambda)_{\text{on}}}; 0 \leq \tau \leq 1$
Transmittance of optical radiation from one direction into another	bidirectional transmittance	lumens	$\tau(\theta_i, \psi_i; \theta_t, \psi_t)$		$\tau(\theta_i, \psi_i; \theta_t, \psi_t) = \frac{\Phi(\theta_t, \psi_t)}{\Phi(\theta_i, \psi_i)}; 0 \leq \tau \leq 1$
Transmitted luminance per unit illuminance of a surface	bidirectional transmittance distribution function	luminance, illuminance	$f_t(\theta_i, \psi_i; \theta_t, \psi_t)$	sr^{-1}	$f_t(\theta_i, \psi_i; \theta_t, \psi_t) = \frac{dL_t(\theta_t, \psi_t)}{E_i(\theta_i, \psi_i)}; 0 \leq f_t < \infty$
Fraction of incident light lost in a material	absorptance	lumens	α		$\alpha \equiv \frac{\Phi_{\text{lost}}}{\Phi_{\text{on}}} = \frac{\Phi_{\text{on}} - \Phi_{\text{out}}}{\Phi_{\text{on}}}; 0 \leq \alpha \leq 1$
Luminous difference of a target and its surround	luminance contrast	luminance	C		$C = \frac{L_t - L_b}{L_b}$, or $C = \left \frac{L_t - L_b}{L_b} \right $, or $C = \frac{L_t - L_b}{L_t + L_b}$
The perception of the luminous strength of luminance	brightness	luminance	B		$B \propto L_t^{1/3} - B_0(L_b, \alpha)$

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